

UNIVERSITY OF KELANIYA - SRI LANKA

FACULTY OF SCIENCE

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Analogue Electronics II
(Operational Amplifiers)

Detailed Model Solutions

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How to use this document

These solutions answer all seven questions, although the examination asks candidates to answer any five. Derivations are arranged in a marking-friendly order: circuit law, substitution, simplification, final result, and physical interpretation. Unless stated otherwise, the op-amp is treated as ideal.

Prepared as a study and revision document

These are model answers, not an official University marking scheme.

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General assumptions, notation, and answering method

For ideal op-amp analysis, use the following facts only when the circuit has **negative feedback** and the output is not saturated:

- Open-loop gain is very large: $A_{OL} \rightarrow \infty$.
- Input resistance is very large: $i_+ = i_- = 0$.
- The differential input voltage is approximately zero: $V_+ \approx V_-$ (virtual short).
- Output resistance is approximately zero.
- For AC-equivalent circuits, ideal DC voltage supplies are replaced by AC ground.

For comparator and Schmitt-trigger operation, the op-amp is deliberately used open-loop or with positive feedback. Therefore, do **not** apply $V_+ = V_-$; instead use

$$V_o = +V_{\text{sat}} \quad \text{if } V_+ > V_-, \quad V_o = -V_{\text{sat}} \quad \text{if } V_+ < V_-.$$

Recommended examination structure

Draw and label the circuit first. State the relevant ideal-op-amp rule. Write the node equation or impedance-divider equation. Simplify symbolically before inserting numerical values. Box the final expression and include units and physical meaning.

Question 1 - Negative feedback and basic closed-loop amplifiers

1(a) Negative-feedback concept

Negative feedback means that a fraction of the output is returned to the input in a polarity that opposes the applied signal. Let A be the op-amp open-loop gain and let a fraction β of the output be fed back.

$$V_f = \beta V_o, \quad V_d = V_s - V_f, \quad V_o = AV_d.$$

Hence,

$$V_o = A(V_s - \beta V_o)$$

$$V_o(1 + A\beta) = AV_s.$$

Therefore, the closed-loop gain is

Closed-loop gain with negative feedback

$$A_f = \frac{V_o}{V_s} = \frac{A}{1 + A\beta}.$$

When $A\beta \gg 1$, $A_f \approx 1/\beta$. Thus, the closed-loop gain is determined mainly by the external feedback network rather than the uncertain open-loop gain.

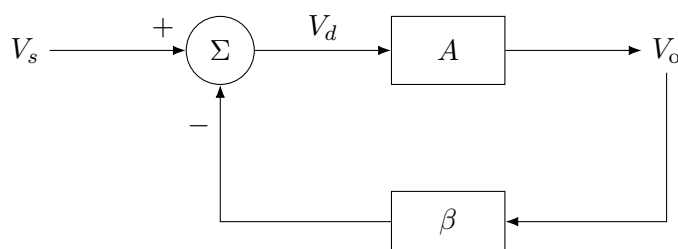


Figure 1: General negative-feedback system.

Important effects are controlled gain, increased bandwidth, reduced distortion, reduced sensitivity to device variations, and improved input/output impedances. Negative feedback does not prevent clipping if the required output exceeds the supply-limited output swing.

1(b) Inverting and non-inverting closed-loop amplifiers

Inverting amplifier

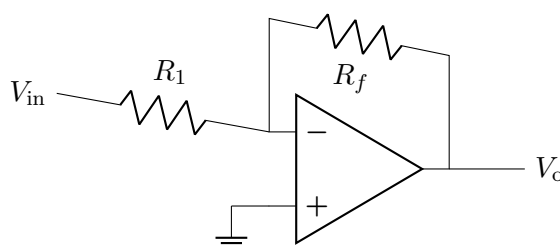


Figure 2: Basic inverting amplifier.

Since $V_+ = 0$, negative feedback gives $V_- \approx 0$. No current enters the input terminal, so KCL at the inverting node gives

$$\frac{V_{in} - 0}{R_1} = \frac{0 - V_o}{R_f}.$$

Therefore,

Inverting amplifier

$$V_o = -\frac{R_f}{R_1} V_{in}, \quad A_{v,inv} = -\frac{R_f}{R_1}.$$

The minus sign represents a 180° phase reversal.

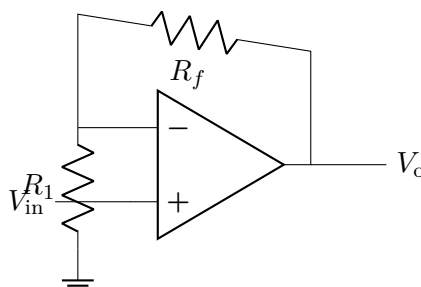
Non-inverting amplifier

Figure 3: Basic non-inverting amplifier.

The feedback-divider voltage is

$$V_- = V_o \frac{R_1}{R_1 + R_f}.$$

Because $V_- = V_+ = V_{in}$,

$$V_{in} = V_o \frac{R_1}{R_1 + R_f}.$$

Therefore,

Non-inverting amplifier

$$V_o = \left(1 + \frac{R_f}{R_1}\right) V_{in}, \quad A_{v,noninv} = 1 + \frac{R_f}{R_1}.$$

The output is in phase with the input.

1(c) Inverter and voltage follower

Unity-gain inverter: select $R_f = R_1$ in the inverting amplifier. Then

$$V_o = -V_{in}, \quad A_v = -1.$$

It changes polarity without changing magnitude.

Voltage follower: connect the output directly to the inverting input and apply the signal to the non-inverting input.

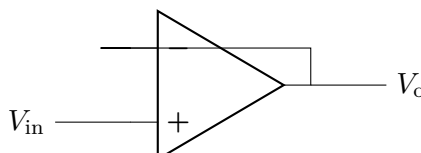


Figure 4: Voltage follower.

Since $V_- = V_o$ and $V_+ = V_{in}$,

$$V_o = V_{in}, \quad A_v = 1.$$

The follower is a buffer: it has very high input resistance, low output resistance, and isolates a weak source from a heavier load.

1(d) Output for switch positions 1, 2, and 3

The three equal $1\text{ k}\Omega$ resistors divide 3 V into equally spaced levels. Since no current enters the non-inverting input, the series $1\text{ k}\Omega$ resistor at the switch does not alter the selected voltage.

$$V_+(1) = 1\text{ V}, \quad V_+(2) = 2\text{ V}, \quad V_+(3) = 3\text{ V}.$$

To obtain the feedback factor, label the inverting-input node $V_A = V_-$ and the central ladder node V_B . Every resistor in the feedback ladder is $1\text{ k}\Omega$.

At node A ,

$$\frac{V_A}{R} + \frac{V_A - V_B}{R} = 0 \Rightarrow V_B = 2V_A.$$

At node B ,

$$\frac{V_B - V_A}{R} + \frac{V_B}{R} + \frac{V_B - V_o}{R} = 0.$$

Thus,

$$3V_B - V_A - V_o = 0.$$

Substituting $V_B = 2V_A$ gives

$$6V_A - V_A = V_o \Rightarrow V_A = \frac{V_o}{5}.$$

During linear negative-feedback operation, $V_A = V_- = V_+$, so

$$\boxed{V_o = 5V_+}.$$

Switch position	V_+	Ideal $5V_+$	Actual output with $V_{\text{sat}} = \pm 10.5\text{ V}$
1	1 V	5 V	$\boxed{5\text{ V}}$
2	2 V	10 V	$\boxed{10\text{ V}}$
3	3 V	15 V	$\boxed{10.5\text{ V}}$ (positive saturation)

Important saturation point

For position 3, the ideal feedback calculation requests 15 V , but the op-amp can produce only 10.5 V . Once saturated, the virtual-short rule is no longer valid.

Question 2 - True and practical differentiators

2(a) Show that the true differentiator differentiates the input

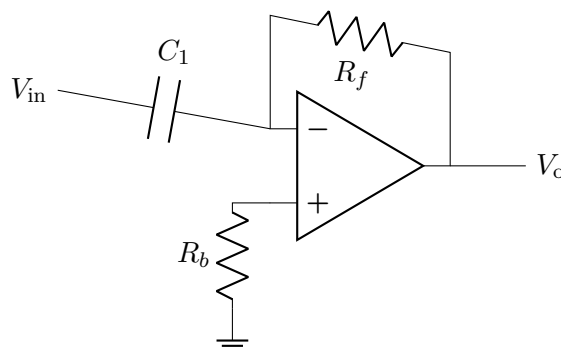


Figure 5: True inverting differentiator. The resistor R_b compensates input-bias-current effects in a practical op-amp.

The non-inverting input is at ground, so $V_- \approx 0$. The capacitor current entering the summing node is

$$i_C = C_1 \frac{d(V_{in} - V_-)}{dt} = C_1 \frac{dV_{in}}{dt}.$$

The feedback current is

$$i_f = \frac{V_- - V_o}{R_f} = -\frac{V_o}{R_f}.$$

KCL gives $i_C = i_f$, hence

Time-domain differentiator equation

$$V_o = -R_f C_1 \frac{dV_{in}}{dt}.$$

The output is proportional to the rate of change of the input and is inverted.

2(b) Transfer function and the high-frequency design problem

Using $Z_C = 1/(sC_1)$,

$$H(s) = \frac{V_o(s)}{V_{in}(s)} = -\frac{R_f}{Z_C} = -sR_f C_1.$$

For $s = j\omega$,

$$|H(j\omega)| = \omega R_f C_1 = 2\pi f R_f C_1, \quad \angle H = -90^\circ.$$

The magnitude increases indefinitely at 20 dB/decade. This causes several practical problems:

- high-frequency noise is strongly amplified;
- capacitor impedance becomes very small, heavily loading the signal source;
- finite op-amp gain-bandwidth and extra phase shift can cause instability or oscillation;
- fast signals may exceed the slew rate or drive the output into saturation.

2(c) Practical differentiator and transfer function

The simplest practical modification is to insert a resistor R_1 in series with C_1 .

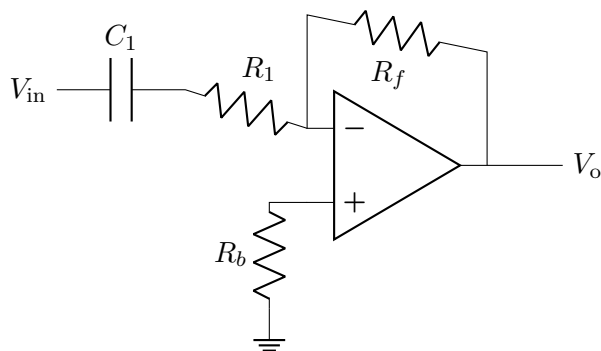


Figure 6: One-pole practical differentiator.

The input impedance is

$$Z_{in} = R_1 + \frac{1}{sC_1}.$$

Therefore,

$$H(s) = -\frac{R_f}{R_1 + 1/(sC_1)} = -\frac{sR_fC_1}{1 + sR_1C_1}.$$

Define

$$\omega_c = \frac{1}{R_1C_1}, \quad f_c = \frac{1}{2\pi R_1C_1}, \quad K = \frac{R_f}{R_1}.$$

Then

Practical differentiator transfer function

$$H(s) = -K \frac{s/\omega_c}{1 + s/\omega_c}.$$

At $f \ll f_c$,

$$H(s) \approx -sR_fC_1,$$

so the circuit behaves approximately as a differentiator. At $f \gg f_c$,

$$H(s) \approx -\frac{R_f}{R_1},$$

so the high-frequency gain is finite rather than unbounded.

Optional enhanced practical differentiator

A capacitor may also be placed in parallel with R_f to introduce an additional high-frequency roll-off. The one-resistor modification above is the first-order form directly required to demonstrate a high-pass response.

2(d) High-pass frequency response

For $s = j\omega$,

$$|H(j\omega)| = \frac{R_f}{R_1} \frac{\omega R_1 C_1}{\sqrt{1 + (\omega R_1 C_1)^2}} = K \frac{f/f_c}{\sqrt{1 + (f/f_c)^2}}.$$

Thus:

- $f \ll f_c$: magnitude rises at 20 dB/decade;
- $f = f_c$: $|H| = K/\sqrt{2}$, which is -3 dB relative to the high-frequency gain;
- $f \gg f_c$: magnitude approaches K .

This is exactly a first-order inverting high-pass response.

2(e) Square-wave input and output waveform

A square wave is constant between its edges, so its derivative is zero there. At a rising edge, dV_{in}/dt is positive, and the inverting differentiator produces a **negative** pulse. At a falling edge, dV_{in}/dt is negative, and the output produces a **positive** pulse.

For a practical differentiator with time constant $\tau = R_1 C_1$, a positive step of size ΔV produces

$$V_o(t) = -\frac{R_f}{R_1} \Delta V e^{-t/\tau}, \quad t \geq 0.$$

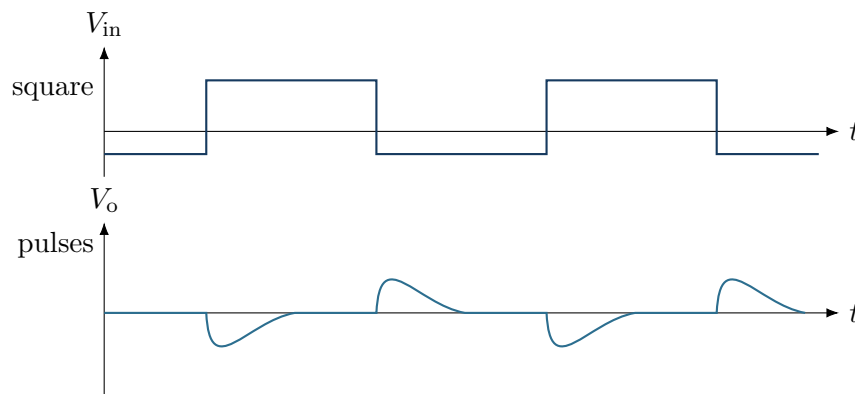


Figure 7: Square-wave differentiation: negative pulses at rising edges and positive pulses at falling edges.

Question 3 - First-order all-pass phase shifter

3(a) Derivation using superposition

The circuit has $R_f = R_1$. The non-inverting input receives the output of an RC low-pass network:

$$V_+ = \frac{1}{1 + sRC} V_{in}.$$

With ideal negative feedback, $V_- = V_+$. KCL at the inverting input gives

$$\frac{V_{in} - V_-}{R_1} + \frac{V_o - V_-}{R_f} = 0.$$

Since $R_f = R_1$,

$$V_{in} + V_o - 2V_- = 0 \Rightarrow V_o = 2V_- - V_{in}.$$

Substitute $V_- = V_{in}/(1 + sRC)$:

$$\frac{V_o}{V_{in}} = \frac{2}{1 + sRC} - 1 = \frac{1 - sRC}{1 + sRC}.$$

With $s = j2\pi f$ and $f_c = 1/(2\pi RC)$,

All-pass transfer function

$$T(jf) = \frac{1 - j(f/f_c)}{1 + j(f/f_c)}, \quad f_c = \frac{1}{2\pi RC}.$$

3(b) Amplitude response and all-pass behaviour

Let $x = f/f_c$. Then

$$|T| = \frac{|1 - jx|}{|1 + jx|} = \frac{\sqrt{1 + x^2}}{\sqrt{1 + x^2}} = 1.$$

Therefore,

$$|T(jf)| = 1 \quad \text{for all frequencies}.$$

All frequencies pass with unity magnitude, while their phase is changed. Hence the circuit is an **all-pass filter**.

3(c) Phase response and phase-shifter type

The numerator angle is $-\tan^{-1} x$ and the denominator angle is $+\tan^{-1} x$. Therefore,

Phase response

$$\phi(f) = -2 \tan^{-1} \left(\frac{f}{f_c} \right).$$

At $f = 0$, $\phi = 0^\circ$. At $f = f_c$, $\phi = -90^\circ$. As $f \rightarrow \infty$, $\phi \rightarrow -180^\circ$. It is therefore a first-order **phase-lag all-pass network**.

3(d) Frequency for a 90° phase shift

For $|\phi| = 90^\circ$,

$$2 \tan^{-1}(f/f_c) = 90^\circ \Rightarrow f = f_c.$$

Given $R = 2 \text{ k}\Omega$ and $C = 0.01 \text{ }\mu\text{F} = 10 \text{ nF}$,

$$f_c = \frac{1}{2\pi(2000)(10 \times 10^{-9})} = 7957.7 \text{ Hz}.$$

$$f \approx 7.96 \text{ kHz}.$$

The value of R_1 does not set f_c ; it must only satisfy $R_f = R_1$ for the stated transfer function.

Question 4 - Inverting Schmitt trigger and oven controller

4(a) Upper and lower threshold voltages

In the given circuit, R_2 connects the output to the non-inverting input and R_1 connects the non-inverting input to ground. The resistor $R_m = R_1 \parallel R_2$ at the inverting input is used mainly to balance input-bias-current resistance and does not affect ideal threshold values.

The positive-feedback factor is

$$\beta = \frac{R_1}{R_1 + R_2}.$$

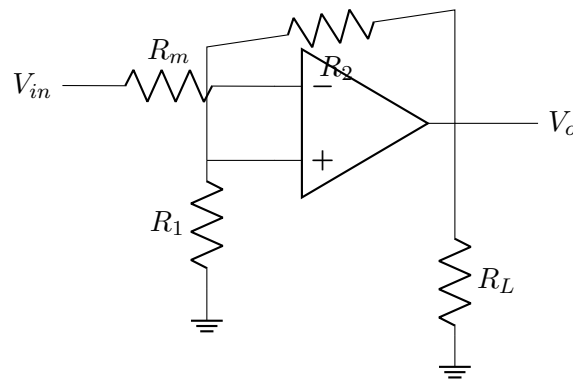


Figure 8: Inverting Schmitt trigger with positive feedback.

Since

$$V_T = V_+ = \beta V_o,$$

when $V_o = +V_{\text{sat}}$,

$$V_{UT} = +\beta V_{\text{sat}},$$

and when $V_o = -V_{\text{sat}}$,

$$V_{LT} = -\beta V_{\text{sat}}.$$

Symmetrical inverting Schmitt-trigger thresholds

$$V_{UT} = +\frac{R_1}{R_1 + R_2} V_{\text{sat}}, \quad V_{LT} = -\frac{R_1}{R_1 + R_2} V_{\text{sat}}.$$

The hysteresis width is

$$V_H = V_{UT} - V_{LT} = 2\beta V_{\text{sat}}.$$

4(b) Operation and voltage-transfer curve

Assume initially that $V_o = +V_{\text{sat}}$. Then $V_+ = +\beta V_{\text{sat}} = V_{UT}$.

1. As V_{in} rises but remains below V_{UT} , $V_+ > V_-$ and the output remains at $+V_{\text{sat}}$.
2. When V_{in} just exceeds V_{UT} , $V_- > V_+$, so the output switches rapidly to $-V_{\text{sat}}$.
3. The threshold immediately changes to $V_{LT} = -\beta V_{\text{sat}}$ because it is derived from the new output level.
4. The output remains at $-V_{\text{sat}}$ while the input decreases until $V_{\text{in}} < V_{LT}$.

5. When V_{in} falls below V_{LT} , $V_+ > V_-$ and the output switches back to $+V_{sat}$.

The different switching points for rising and falling inputs create hysteresis, which rejects small noise near a single threshold and prevents output chatter.

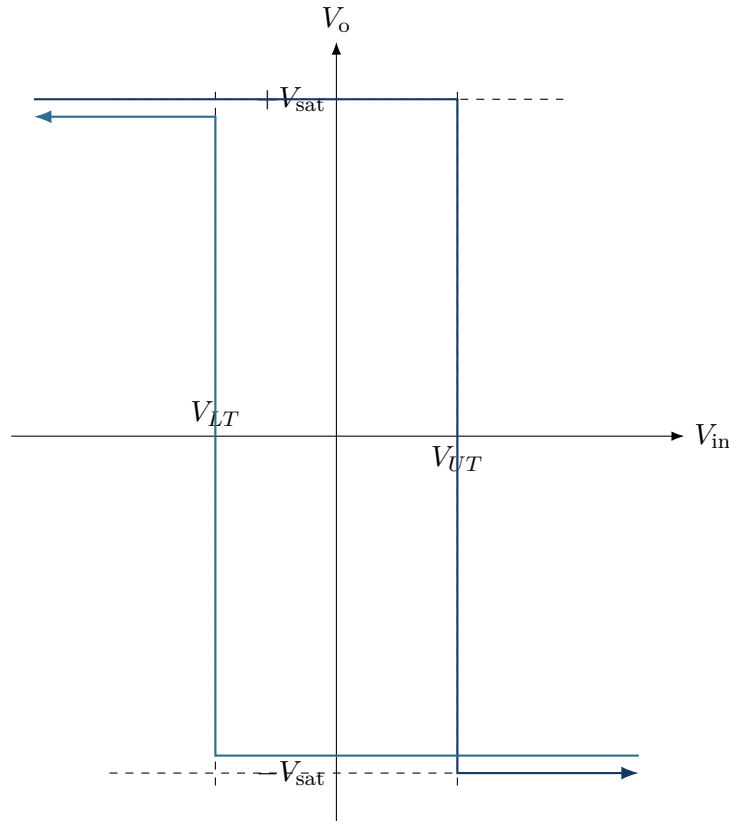


Figure 9: Voltage-transfer characteristic of an inverting Schmitt trigger.

4(c) Shift both trip points in the positive direction

Replace the ground connection at the lower end of R_1 with a positive DC reference V_R . The non-inverting input is then the conductance-weighted average of V_o and V_R :

$$V_+ = \frac{R_1 V_o + R_2 V_R}{R_1 + R_2}.$$

Define

$$\beta = \frac{R_1}{R_1 + R_2}, \quad \gamma = \frac{R_2}{R_1 + R_2} = 1 - \beta.$$

Then

$$V_+ = \beta V_o + \gamma V_R.$$

Therefore,

Positively shifted thresholds

$$V_{UT} = \beta V_{sat} + \gamma V_R, \quad V_{LT} = -\beta V_{sat} + \gamma V_R.$$

The hysteresis width remains $2\beta V_{sat}$, while its centre shifts from zero to

$$V_C = \frac{V_{UT} + V_{LT}}{2} = \gamma V_R.$$

A positive V_R shifts both thresholds in the positive direction.

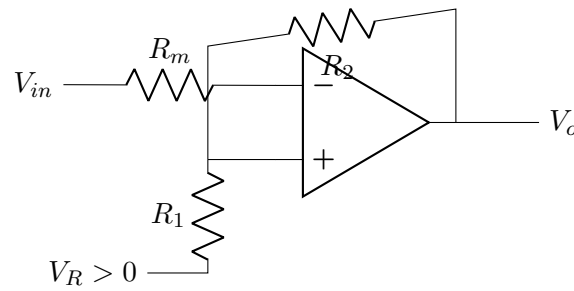


Figure 10: Positive-reference modification that shifts both trip points upward.

4(d) Industrial oven controller for 200 °C to 240 °C

Let the sensor output be proportional to temperature:

$$V_S = kT,$$

where k is the sensor sensitivity in volts per degree Celsius. The desired sensor thresholds are

$$V_L = 200k, \quad V_U = 240k.$$

Use the positively shifted **inverting** Schmitt trigger with the sensor connected to the inverting input. Arrange the power driver so that a positive op-amp output energises a relay or transistor and turns the heater ON.

Because the output saturation values are ± 13.5 V,

$$V_U - V_L = 2\beta V_{\text{sat}}.$$

Thus,

General design equations

$$\beta = \frac{V_U - V_L}{2V_{\text{sat}}} = \frac{40k}{27},$$

$$\frac{R_1}{R_1 + R_2} = \beta \Rightarrow \frac{R_2}{R_1} = \frac{1 - \beta}{\beta},$$

and because the threshold centre is $(V_U + V_L)/2 = 220k$,

$$V_R = \frac{220k}{1 - \beta}.$$

Why a unique numerical answer is not possible from the paper alone

The paper states only that the sensor voltage is proportional to temperature; it does not state the sensitivity k . Therefore, resistor and reference values must either be expressed in terms of k or calculated after choosing a real sensor.

Illustrative design using a 10 mV/°C sensor

For a sensor such as an LM35-type device, take $k = 10$ mV/°C. Then

$$V_L = 2.00 \text{ V}, \quad V_U = 2.40 \text{ V}.$$

The required feedback fraction is

$$\beta = \frac{2.40 - 2.00}{2(13.5)} = 0.014815.$$

Hence

$$\frac{R_2}{R_1} = \frac{1 - 0.014815}{0.014815} = 66.5.$$

Choose

$$R_1 = 1.00 \text{ k}\Omega, \quad R_2 = 66.5 \text{ k}\Omega$$

(or a nearby standard value plus a trim resistor). The required reference is

$$V_R = \frac{2.20}{1 - 0.014815} = 2.233 \text{ V}.$$

Choose $R_m \approx R_1 \parallel R_2 \approx 985 \Omega$, so $1 \text{ k}\Omega$ is suitable.

Control sequence:

- Below 200°C , $V_S < V_L$, the output switches to $+13.5 \text{ V}$ and the heater turns ON.
- The heater remains ON while temperature rises through the band.
- Above 240°C , $V_S > V_U$, the output switches to -13.5 V and the heater turns OFF.
- It remains OFF until the temperature again falls below 200°C .

Power-interface requirement

The heating element must not be connected directly to the op-amp. Use a transistor or MOSFET driver and a suitably rated relay/contact or solid-state relay. Add a flyback diode across a DC relay coil.

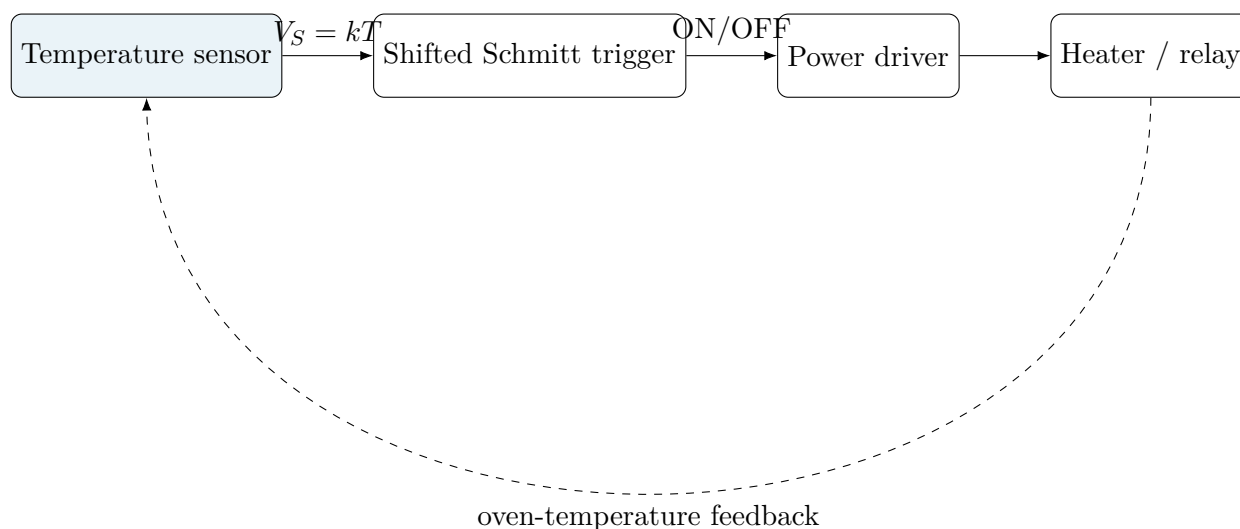


Figure 11: Closed-loop oven-temperature control arrangement.

Question 5 - Passive and active first-order filters

5(a) Passive RC low-pass and high-pass filters

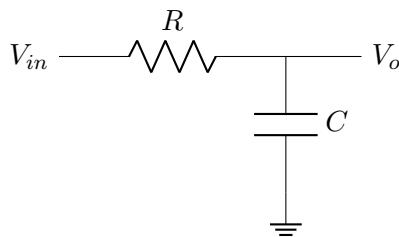
Passive low-pass filter

Figure 12: First-order passive RC low-pass filter.

The output is taken across the capacitor. With $Z_C = 1/(sC)$,

$$H_{LP}(s) = \frac{Z_C}{R + Z_C} = \frac{1}{1 + sRC}.$$

Thus,

$$H_{LP}(s) = \frac{1}{1 + sRC}, \quad \omega_c = \frac{1}{RC}, \quad f_c = \frac{1}{2\pi RC}.$$

At low frequency, $H \approx 1$; at high frequency, the magnitude falls at 20 dB/decade.

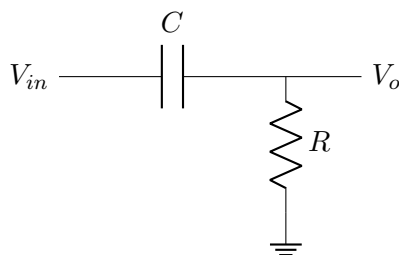
Passive high-pass filter

Figure 13: First-order passive RC high-pass filter.

The output is taken across the resistor:

$$H_{HP}(s) = \frac{R}{R + 1/(sC)} = \frac{sRC}{1 + sRC}.$$

Therefore,

$$H_{HP}(s) = \frac{sRC}{1 + sRC}, \quad f_c = \frac{1}{2\pi RC}.$$

At low frequency, the capacitor blocks the signal; at high frequency, the output approaches the input.

5(b) Active non-inverting low-pass and high-pass filters

Place a non-inverting op-amp after each RC network. Its high input resistance avoids significant loading. Let

$$K = 1 + \frac{R_f}{R_g}.$$

Then

Active first-order responses

$$H_{LP,a}(s) = \frac{K}{1 + sR_L C_L},$$

$$H_{HP,a}(s) = K \frac{sR_H C_H}{1 + sR_H C_H}.$$

The RC cut-off frequencies are unchanged; the op-amp adds gain and buffering.

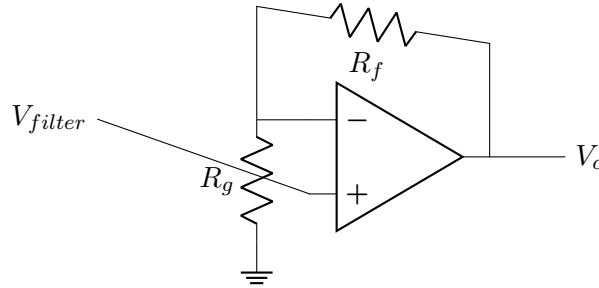


Figure 14: Non-inverting gain stage placed after either passive RC network.

5(c) Active non-inverting band-pass filter with single-stage gain control

A wide-band first-order band-pass can be formed by cascading a passive high-pass section and passive low-pass section, followed by one non-inverting op-amp gain stage.

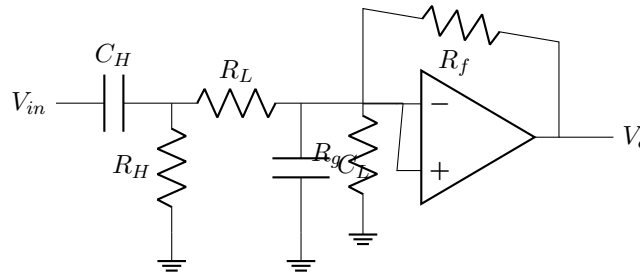


Figure 15: Single-op-amp active band-pass filter. Choose $R_L \gg R_H$ to reduce interaction between the passive sections.

Assuming the sections are sufficiently impedance-scaled so that loading is small,

Approximate band-pass transfer function

$$H_{BP}(s) = K \left(\frac{sR_H C_H}{1 + sR_H C_H} \right) \left(\frac{1}{1 + sR_L C_L} \right)$$

$$H_{BP}(s) = K \frac{sR_H C_H}{(1 + sR_H C_H)(1 + sR_L C_L)}.$$

The approximate corner frequencies are

$$f_L = \frac{1}{2\pi R_H C_H}, \quad f_H = \frac{1}{2\pi R_L C_L}, \quad f_L < f_H.$$

At low frequency the response rises at 20 dB/decade; in the pass band it is approximately K ; above f_H it falls at 20 dB/decade.

For an exact unbuffered passive cascade, loading adds an extra term. If the high-pass is followed directly by the low-pass,

$$\frac{V_+}{V_{in}} = \frac{sR_H C_H}{1 + s(R_H C_H + R_L C_L + R_H C_L) + s^2 R_H R_L C_H C_L}.$$

Selecting $R_L \gg R_H$ makes the loading term $R_H C_L$ relatively small.

5(d) Design for a 1 kHz to 10 kHz pass band and gain 2

Required:

$$f_L = 1 \text{ kHz}, \quad f_H = 10 \text{ kHz}, \quad K = 2.$$

For the high-pass section, choose $C_H = 100 \text{ nF}$:

$$R_H = \frac{1}{2\pi f_L C_H} = \frac{1}{2\pi(1000)(100 \times 10^{-9})} = 1.59 \text{ k}\Omega.$$

Use $R_H = 1.6 \text{ k}\Omega$ and $C_H = 100 \text{ nF}$, giving approximately 995 Hz.

For the low-pass section, choose $C_L = 1 \text{ nF}$:

$$R_L = \frac{1}{2\pi f_H C_L} = \frac{1}{2\pi(10^4)(10^{-9})} = 15.9 \text{ k}\Omega.$$

This gives $R_L/R_H \approx 10$, which helps reduce loading.

For gain 2,

$$2 = 1 + \frac{R_f}{R_g} \Rightarrow R_f = R_g.$$

Choose $R_f = R_g = 10 \text{ k}\Omega$.

Final practical design

$$R_H = 1.6 \text{ k}\Omega, \quad C_H = 100 \text{ nF}$$

$$R_L = 15.9 \text{ k}\Omega, \quad C_L = 1 \text{ nF}$$

$$R_f = R_g = 10 \text{ k}\Omega \quad (K = 2).$$

Question 6 - Wien-bridge network and oscillator

6(a) Transfer function and resonant frequency

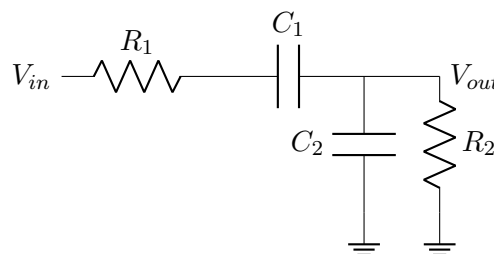


Figure 16: Wien lead-lag network from the examination paper.

For the network in the paper, the series arm is

$$Z_1 = R_1 + \frac{1}{sC_1} = \frac{1 + sR_1 C_1}{sC_1},$$

and the shunt arm is

$$Z_2 = R_2 \parallel \frac{1}{sC_2} = \frac{R_2}{1 + sR_2C_2}.$$

The output is taken across Z_2 , so

$$H(s) = \frac{Z_2}{Z_1 + Z_2}.$$

Substitution and simplification give

General Wien-network transfer function

$$H(s) = \frac{sR_2C_1}{1 + s(R_1C_1 + R_2C_2 + R_2C_1) + s^2R_1R_2C_1C_2}.$$

For $R_1 = R_2 = R$ and $C_1 = C_2 = C$,

$$H(s) = \frac{sRC}{1 + 3sRC + s^2R^2C^2}.$$

Set $s = j\omega$:

$$H(j\omega) = \frac{j\omega RC}{1 - \omega^2R^2C^2 + j3\omega RC}.$$

The phase is zero when the real part of the denominator is zero:

$$1 - \omega_0^2R^2C^2 = 0.$$

Therefore,

Resonant frequency and attenuation

$$\omega_0 = \frac{1}{RC}, \quad f_0 = \frac{1}{2\pi RC}.$$

At $\omega_0 RC = 1$,

$$H(j\omega_0) = \frac{1}{3}.$$

6(b) Wien-bridge oscillator and conditions for oscillation

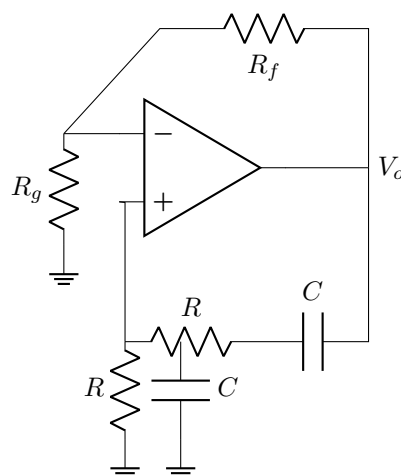


Figure 17: Op-amp Wien-bridge oscillator with equal R and equal C .

Connect the Wien network from the op-amp output to the non-inverting input. Use a resistive negative-feedback divider at the inverting input to set the non-inverting amplifier gain

$$A = 1 + \frac{R_f}{R_g}.$$

At f_0 , the Wien network contributes zero phase shift and attenuation $\beta = 1/3$.

The Barkhausen conditions are

$$|A\beta| = 1, \quad \angle(A\beta) = 0^\circ + 360^\circ n.$$

Therefore,

$$A \left(\frac{1}{3} \right) = 1 \quad \Rightarrow \quad \boxed{A = 3}.$$

For the non-inverting amplifier,

$$1 + \frac{R_f}{R_g} = 3 \quad \Rightarrow \quad \boxed{R_f = 2R_g}.$$

For reliable start-up, the initial gain is commonly set slightly above 3. An automatic amplitude-control element (small lamp, diodes, JFET, or AGC circuit) then reduces the effective gain to 3 as the sine-wave amplitude builds. If the final gain remains above 3, the output clips; if it is below 3, oscillations decay.

6(c) Design for 1 kHz

Choose equal capacitors

$$C_1 = C_2 = 10 \text{ nF}.$$

Then

$$R = \frac{1}{2\pi f_0 C} = \frac{1}{2\pi(1000)(10 \times 10^{-9})} = 15.915 \text{ k}\Omega.$$

Use precision values $R_1 = R_2 = 15.9 \text{ k}\Omega$. This gives

$$f_0 \approx \frac{1}{2\pi(15.9 \times 10^3)(10 \times 10^{-9})} = 1001 \text{ Hz}.$$

For gain 3, choose

$$R_g = 10 \text{ k}\Omega, \quad R_f = 20 \text{ k}\Omega.$$

One valid 1 kHz design

$$\boxed{R_1 = R_2 = 15.9 \text{ k}\Omega, \quad C_1 = C_2 = 10 \text{ nF}}$$

$$\boxed{R_g = 10 \text{ k}\Omega, \quad R_f = 20 \text{ k}\Omega}.$$

Use 1% resistors and stable film capacitors for good frequency accuracy.

Question 7 - Op-amp voltage regulators

7(a) Line regulation and load regulation

Line regulation measures how well a regulator holds its output constant when the input supply changes while the load is fixed:

$$\text{Line regulation} = \frac{\Delta V_{out}}{\Delta V_{in}}$$

often expressed in mV/V or as a percentage.

Load regulation measures how well the output remains constant when load current changes while input voltage is fixed:

$$\text{Load regulation} = V_{NL} - V_{FL}$$

or

$$\% \text{ load regulation} = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%.$$

Here V_{NL} is the no-load voltage and V_{FL} is the full-load voltage. Smaller values indicate better regulation.

7(b) Basic op-amp series regulator

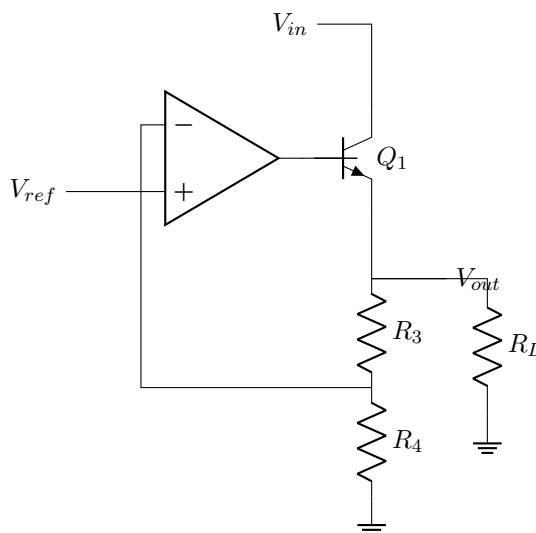


Figure 18: Basic op-amp series regulator using an NPN pass transistor.

A basic series regulator contains:

- a stable reference voltage V_Z or V_{ref} ;
- an output sampling divider;
- an op-amp error amplifier;
- a power transistor in series between the unregulated input and load.

Let R_3 be the upper divider resistor from V_{out} to the feedback node and R_4 the lower resistor to ground. The sampled voltage is

$$V_s = V_{out} \frac{R_4}{R_3 + R_4}.$$

Negative feedback drives the pass transistor until $V_s \approx V_{ref}$. Hence,

Series-regulator output voltage

$$V_{out} = V_{ref} \left(1 + \frac{R_3}{R_4} \right).$$

If V_{out} falls because the load current increases, V_s falls below V_{ref} , the op-amp increases the pass-transistor drive, and V_{out} is restored. If V_{out} rises, the op-amp reduces drive. The pass element carries approximately the load current and dissipates

$$P_Q \approx (V_{in} - V_{out})I_L.$$

7(c) Series regulator with constant-current limiting

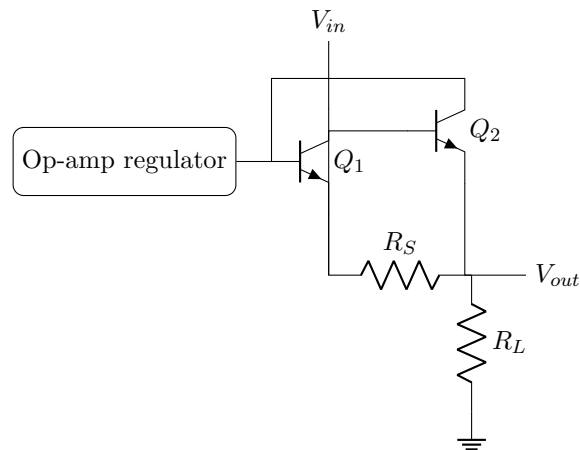


Figure 19: Conceptual constant-current limiter. Q_2 senses the voltage across R_S and removes drive from Q_1 .

Insert a small current-sense resistor R_S in series with the load path. Connect a limiting transistor Q_2 so that it senses the voltage across R_S and steals or clamps the base drive of the series pass transistor Q_1 .

For normal current,

$$I_L R_S < V_{BE(on)},$$

so Q_2 is OFF and the voltage-control loop regulates normally.

When

$$I_L R_S \approx V_{BE(on)} \approx 0.6 \text{ V to } 0.7 \text{ V},$$

Q_2 turns ON and prevents any substantial increase in Q_1 drive. Therefore,

Constant-current limit

$$I_{L,\max} \approx \frac{V_{BE(on)}}{R_S}.$$

For example, using 0.65 V, $R_S = 0.65 \Omega$ gives approximately 1 A.

In current limiting, the output voltage falls as necessary to keep the current near the set value. The pass transistor must be checked for worst-case safe-operating-area and power dissipation, especially during a short circuit.

7(d)(i) Required power rating of R_1 in Figure 6

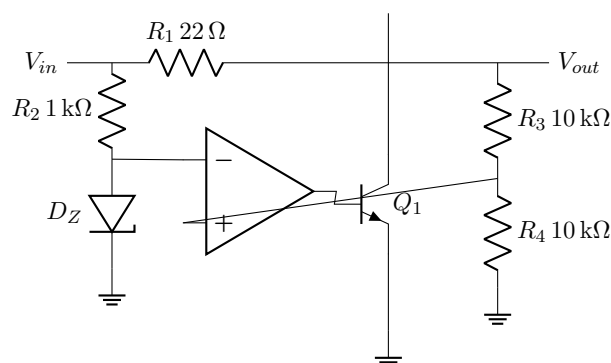


Figure 20: Redrawn basic op-amp shunt regulator of Figure 6.

In the shunt regulator, $R_1 = 22\ \Omega$ is the series current-limiting resistor between V_{in} and the regulated output node. The maximum resistor dissipation occurs in the worst case when the output node is pulled close to ground, so nearly the full maximum input voltage appears across R_1 :

$$P_{R_1, \max} = \frac{V_{in, \max}^2}{R_1} = \frac{(12.5)^2}{22} = 7.10\ \text{W}.$$

Required resistor rating

$$P_{R_1} \geq 7.1\ \text{W}.$$

The next safe standard rating is typically

$$10\ \text{W}.$$

The resistor must also tolerate its temperature rise and should be mounted with suitable clearance from heat-sensitive components.

7(d)(ii) Main drawback of the shunt regulator

The shunt transistor is connected across the output and continuously diverts the current not required by the load. The series resistor therefore carries

$$I_{R_1} = I_L + I_{shunt}$$

and can dissipate several watts even when the load requires little current.

Main comparison

The main drawback is **poor efficiency and high continuous power dissipation**. A shunt regulator wastes current in the shunt element and the series resistor, whereas a series regulator supplies approximately only the load current through its pass element.

Other consequences are the need for high-wattage components, additional heat, and limited suitability for high-current loads. Its advantages are simplicity, fast response, and natural current limitation by the series resistor.

Final revision summary

Topic	Key result
Negative feedback	$A_f = A/(1 + A\beta) \approx 1/\beta$ when $A\beta \gg 1$.
Inverting amplifier	$V_o = -(R_f/R_1)V_{in}$.
Non-inverting amplifier	$V_o = (1 + R_f/R_1)V_{in}$.
Q1 ladder circuit	Feedback factor $\beta = 1/5$; outputs are 5 V, 10 V, and 10.5 V saturated.
True differentiator	$V_o = -R_f C dV_{in}/dt$, $H(s) = -sR_f C$.
Practical differentiator	$H(s) = -sR_f C/(1 + sR_1 C)$; first-order high-pass, bounded HF gain.
All-pass phase shifter	$T = (1 - sRC)/(1 + sRC)$, $ T = 1$, $\phi = -2 \tan^{-1}(f/f_c)$.
Schmitt trigger	$V_{UT} = +\beta V_{sat}$, $V_{LT} = -\beta V_{sat}$, $\beta = R_1/(R_1 + R_2)$.
Shifted Schmitt trigger	$V_{UT} = \beta V_{sat} + (1 - \beta)V_R$, $V_{LT} = -\beta V_{sat} + (1 - \beta)V_R$.
Passive low-pass	$H = 1/(1 + sRC)$.
Passive high-pass	$H = sRC/(1 + sRC)$.
Filter corner	$f_c = 1/(2\pi RC)$.
Wien network	$f_0 = 1/(2\pi RC)$, $\beta(f_0) = 1/3$.

Topic	Key result
Wien oscillator	Amplifier gain $A = 3$, so $R_f = 2R_g$.
Series regulator	$V_{out} = V_{ref}(1 + R_3/R_4)$.
Current limit	$I_{max} \approx V_{BE}/R_S$.
Figure 6 resistor	$P_{R_1, max} = 12.5^2/22 = 7.10 \text{ W}$; use a 10 W part.

Final check before the examination

Practise drawing each circuit from memory, and do not apply the virtual-short rule to comparators, Schmitt triggers, or saturated op-amps. In numerical designs, show the formula, substitution, calculated value, selected standard component value, and the resulting practical frequency or threshold.